Exercise session n°3 : stochastic bandits (part 2) and KL divergence

Exercise 1 :

Consider the ε-greedy algorithm with $\varepsilon_t = \min\left(1, \frac{(K\ln(t))^{\frac{1}{3}}}{t^{\frac{1}{3}}}\right)$ for any $t \in \mathbb{N}$. Show that for a large enough universal constant $C > 0$, the regret of ε-greedy satisfies

$$R_T \leq CT^{\frac{2}{3}}(K\ln(T))^{\frac{1}{3}}.$$

Hint: Bound the instantaneous regret $\mathbb{E}[\Delta_{a_t}]$.

Exercise 2 :

Concentration for sequences of random length. Let $X_1, X_2, \ldots$ be a sequence of independent standard Gaussian random variables defined on probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Suppose that $T : \Omega \to \{1, 2, 3, \ldots\}$ is another variable and let $\hat{\mu}_T = \sum_{t=1}^{T} \frac{X_t}{T}$ be the empirical mean based on T samples.

1) Show that if T is independent from X_t for all t, then for any $\delta \in (0, 1)$

$$\mathbb{P}\left(\hat{\mu}_T \geq \sqrt{\frac{2\ln(1/\delta)}{T}}\right) \leq \delta.$$

2) Now relax the assumption that T is independent from $(X_t)_t$. Let $E_t = \mathbb{1}_{T=t}$ and $\mathcal{F}_t = \sigma(X_1, \ldots, X_t)$ be the σ-algebra generated by the first t samples. Let $\delta \in (0, 1)$ and show there exists a random variable T such that for all t, E_t is $\mathcal{F}_t$-measurable and

$$\mathbb{P}\left(\hat{\mu}_T \geq \sqrt{\frac{2\ln(1/\delta)}{T}}\right) = 1.$$

Hint: You can use the law of the iterated logarithm, which says if $X_1, X_2, \ldots$ is a sequence of independent and identically distributed random variables with zero mean and unit variance, then

$$\limsup_{n\to\infty} \frac{\sum_{t=1}^{n} X_t}{\sqrt{2n\ln\ln n}} = 1 \quad \text{almost surely.}$$

3) What is the relation between the above inequality and our concentration lemma for the empirical means in bandits problems? Do **2)** and our lemma contradict? Why?

Exercise 3 :

Phased SE. Consider the following phased Successive Eliminations algorithm parameterized by $a > 1$. **1)** Show a regret bound similar to Successive Eliminations algorithm.

2) What is the role played by a?

Algorithm: Phased Successive Eliminations
input: T, $a \geq 1$
$\mathcal{K} \leftarrow [K]$
$\ell \leftarrow 0$
while $\mathrm{Card}(\mathcal{K}) > 1$ **do**
  pull all arms in $\mathcal{A}$ $\lceil a^\ell \rceil$ times
  for *all* $k \in \mathcal{K}$ *such that* $\hat{\mu}_k + \sqrt{\frac{2\ln T}{N_k(T)}} \leq \max_{i\in\mathcal{K}} \hat{\mu}_i - \sqrt{\frac{2\ln T}{N_i(T)}}$ **do** $\mathcal{A} \leftarrow \mathcal{A} \setminus \{k\}$
  $\ell \leftarrow \ell + 1$
repeat pull only arm in $\mathcal{K}$ **until** $t = T$

Exercise 4 :

Adapting to reward variance. Let $X_1, \ldots, X_N$ be a sequence of i.i.d. random variables with mean μ, variance σ^2 and bounded support so that $X_t \in [0, M]$ almost surely. Define the estimators

$$\hat{\mu}_N = \frac{1}{N}\sum_{t=1}^{N} X_t$$
$$\hat{\sigma}^2 = \frac{1}{N}\sum_{t=1}^{N}(\hat{\mu} - X_t)^2.$$

We admit in the following the **empirical Bernstein** inequality:

$$\mathbb{P}\left(|\hat{\mu}_N - \mu| \geq \sqrt{\frac{2\hat{\sigma}^2}{N}\ln(3/\delta)} + \frac{3M}{N}\ln(3/\delta)\right) \leq \delta.$$

1) Show that the Bernstein inequality in the course implies here

$$\mathbb{P}\left(|\hat{\mu}_N - \mu| \geq \sqrt{\frac{2\sigma^2}{N}\ln(2/\delta)} + \frac{2M}{3N}\ln(2/\delta)\right) \leq \delta$$

Comment on the differences between the two above *Bernstein inequalities*.

2) Show that $\hat{\sigma}^2 = \frac{1}{N}\sum_{t=1}^{N}(X_t - \mu)^2 - (\hat{\mu}_N - \mu)^2$.

3) Is $\hat{\sigma}^2$ an unbiased estimator of σ^2? If not, can we easily make it unbounded?

4) Show that

$$\mathbb{P}\left(\hat{\sigma}^2 \geq \sigma^2 + \sqrt{\frac{2M^2\sigma^2}{N}\ln(1/\delta)} + \frac{2M^2}{3N}\ln(1/\delta)\right) \leq \delta.$$

Hint: Use Bernstein inequality of **1)**.

5) **(Hard)** Consider a bandit setting with K arms, bounded rewards $X_k(t) \in [0, M]$ and the variance of the k-th arm is σ_k^2. Design a policy that depends on M, but does not need to know σ_i a priori, such that there exists a universal constant $C > 0$ with

$$R_T \leq C \sum_{k, \Delta_k > 0} \left(\Delta_k + (M + \frac{\sigma_i^2}{\Delta_i}) \ln T \right).$$

Hint: Without a complete proof, the empirical Bernstein inequality can be extended (up to some changes) to cases where N is a random variable.